

5. (25 marks) Consider the following quadratic form

$$Q(\mathbf{x}) = bx_1^2 + a(x_2^2 + x_3^2) + 2bx_2x_3.$$

- (a) (4 marks) Identify the *symmetric matrix of coefficients* A associated with this quadratic form. That is

$$Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$$

Find all the principal submatrices of A .

- (b) (7 marks) Define what it means for the quadratic form $Q(\mathbf{x})$ to be *indefinite*. How would you be able to assess, in terms of the properties of A , whether $Q(\mathbf{x})$ is indefinite?
- (c) (14 marks) Find the values of the parameter values, a and b , such that the quadratic form $Q(\mathbf{x})$ is respectively *indefinite*, *positive semidefinite* and *negative semidefinite*? Then, plot these regions in $\mathbb{R}^2$.

6. (25 marks) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a *quasi-concave* function and $g : \mathbb{R} \rightarrow \mathbb{R}$ be a *weakly increasing* function,

$$x > y \Rightarrow g(x) \geq g(y).$$

Let $h(x) = g(f(x))$ for all $x \in \mathbb{R}$.

- (a) (7 marks) Provide two equivalent definitions of *quasi-concavity*.
- (b) (10 marks) Is $h : \mathbb{R} \rightarrow \mathbb{R}$ *quasi-concave*? Prove your answer.
- (c) (8 marks) If x^* maximizes h , does it also maximize f ? If x^* maximizes f , does it also maximize h ?

7. (50 marks) For $a \in (0, \infty)$, consider the problem:

$$\begin{aligned} \max_{x,y} \quad & ax + y \\ \text{s.t.} \quad & x^2 + y^2 \leq 1 \\ & y \geq ax, x \geq 0. \end{aligned}$$

- (a) (4 marks) Establish whether the objective function and the constraints are respectively concave, convex, or neither.
- (b) (4 marks) Can you solve this optimization problem using the Kuhn-Tucker method? That is, for what values of $a > 0$ is the Slater condition satisfied?
- (c) (8 marks) Using the Kuhn-Tucker approach, write down the necessary first order conditions that must be satisfied by the solution of the constrained optimization problem. Are solutions to these conditions also maximizers of the Lagrangean?
- (d) (18 marks) Solve the constrained optimization problem in terms of a .
- (e) (10 marks) Exploit the Envelope Theorem to find the slope of value function with respect to a . State the result and apply it to this setting.
- (f) (6 marks) For what value of $a \in [0, 10]$ is the value function maximized?